Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 3

Full Marks: 20 *Deadline: 11:59 PM, 07-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Execute all the commands in Linux and take the screenshots.
3) Prepare a PDF that contains the question number and corresponding screenshots of the outputs.
4) The naming of the file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, your roll number is B24CS055, then the filename should be:
   B24CS055_Shareen_03.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Getting familiar with piping.
2) Hands-on experience of the effect of executing two or more commands with piping.
3) Getting experience with network commands in Linux.
4) Implementing your own Linux command in the C programming language.

Task 1: Piping [10 Marks]

[Instruction: If the file is being modified, you have to show the contents of the file with the cat command.]

Read the whole question carefully:

Create a directory named activity and enter into the directory. Create a file named emp_data.txt with the following data:

Employee ID	Name	Department	Position	Salary (INR)
101	Anjali Sharma	HR	Manager	8,50,000
102	Ravi Verma	IT	Software Engineer	7,20,000
103	Priya Desai	Marketing	Executive	6,00,000
104	Amitabh Roy	Finance	Analyst	7,80,000
105	Sneha Iyer	IT	QA Tester	5,50,000
106	Karan Mehta	HR	Recruiter	5,75,000
107	Neha Kulkarni	IT	DevOps Engineer	8,20,000
108	Arjun Singh	IT	Manager	9,00,000
109	Ritu Gupta	Finance	Accountant	6,50,000
110	Suresh Patil	IT	Tech Lead	10,00,000

You should create the file with the following format:

101,Anjali Sharma,HR,Manager,850000

…

110,Suresh Patil,IT,Tech Lead,1000000

Now execute the following instructions on the data:

1. Sort data in descending order of the salary and append the records from the sorted data whose salary is greater than 6 lakhs to a new file named senior_emp.txt by replacing ‘,’ with ‘:’ as field separator. You are only allowed to use piping. [2 Marks]

2. Now sort the records of senior_emp.txt with respect to employee ID in increasing order and print only employee records who work in the IT department, except the first two employees of the IT department whose ID number is numerically less than the others. During this execution, you should not modify or create any other file. Print the senior_emp.txt file contents after this execution to verify you have not modified the file. You are allowed to use piping only.

[2 Marks]

3. Now, from emp_data.txt, print the employee ID, employee name, and employee position whose salary is highest among the employees whose name starts with "A" in the format = ID::Name::Position. [3 Marks]
4. Delete the record of the employee from the IT department whose salary is the second highest, and the emp_data.txt should be updated with the new modified content.

[3 Marks]

Task 2: Network Command and Designing Own Command [10 Marks]

1. Use the Linux command to show your machine's IP address. [1 Mark]
2. Show the IP address of iitbhilai.ac.in and do a ping test on it [2 Marks]
3. Write a code in C to match the occurrence of a substring in a file.
   Input Syntax: ./executable_programfile <substring> <filename>
   Output Syntax: <number of occurrences of substring> <file_name> [7 Marks]